

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(trend)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(Kendall)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
#Set theme for plots
```

```
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
```

```
# Set working directory and read the data individually
```

```
Ozone_data_2010 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2011 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2012 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2013 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2014 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2015 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2016 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv",
  stringsAsFactors = FALSE)

Ozone_data_2017 <- read.csv("../Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv",
```

```

stringsAsFactors = FALSE)

Ozone_data_2018 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv",
stringsAsFactors = FALSE)

Ozone_data_2019 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv",
stringsAsFactors = FALSE)

getwd()

```

```
## [1] "/home/guest/EDE_FALL 2025"
```

```

# Put everything together
GaringerAlldata2010_2019_raw <- bind_rows(Ozone_data_2010, Ozone_data_2011, Ozone_data_2012,
Ozone_data_2013, Ozone_data_2014, Ozone_data_2015,
Ozone_data_2016, Ozone_data_2017, Ozone_data_2018,
Ozone_data_2019)

# Check dimensions
dim(GaringerAlldata2010_2019_raw)

```

```
## [1] 3589 20
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```

# 3.
# Date column as date class
GaringerAlldata2010_2019_raw$Date <- mdy(GaringerAlldata2010_2019_raw$Date)

print("Date conversion check:")

```

```
## [1] "Date conversion check:"
```

```
class(GaringerAlldata2010_2019_raw$Date)
```

```
## [1] "Date"
```

```
head(GaringerAlldata2010_2019_raw$Date)
```

```
## [1] "2010-01-01" "2010-01-02" "2010-01-03" "2010-01-04" "2010-01-05"  
## [6] "2010-01-07"
```

```
sum(is.na(GaringerAlldata2010_2019_raw$Date))
```

```
## [1] 0
```

```
# 4.  
GaringerOzone <- GaringerAlldata2010_2019_raw %>%  
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)  
  
# 5.  
Days <- as.data.frame(seq(as.Date("2010-01-01"), as.Date("2019-12-31"), by = "day"))  
names(Days) <- "Date"  
  
# 6.  
GaringerOzone <- Days %>%  
  left_join(GaringerOzone, by = "Date")  
  
# Check data dimensions  
dim(GaringerOzone)
```

```
## [1] 3652    3
```

```
head(GaringerOzone)
```

```
##           Date Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE  
## 1 2010-01-01                                0.031             29  
## 2 2010-01-02                                0.033             31  
## 3 2010-01-03                                0.035             32  
## 4 2010-01-04                                0.031             29  
## 5 2010-01-05                                0.027             25  
## 6 2010-01-06                                NA              NA
```

Visualize

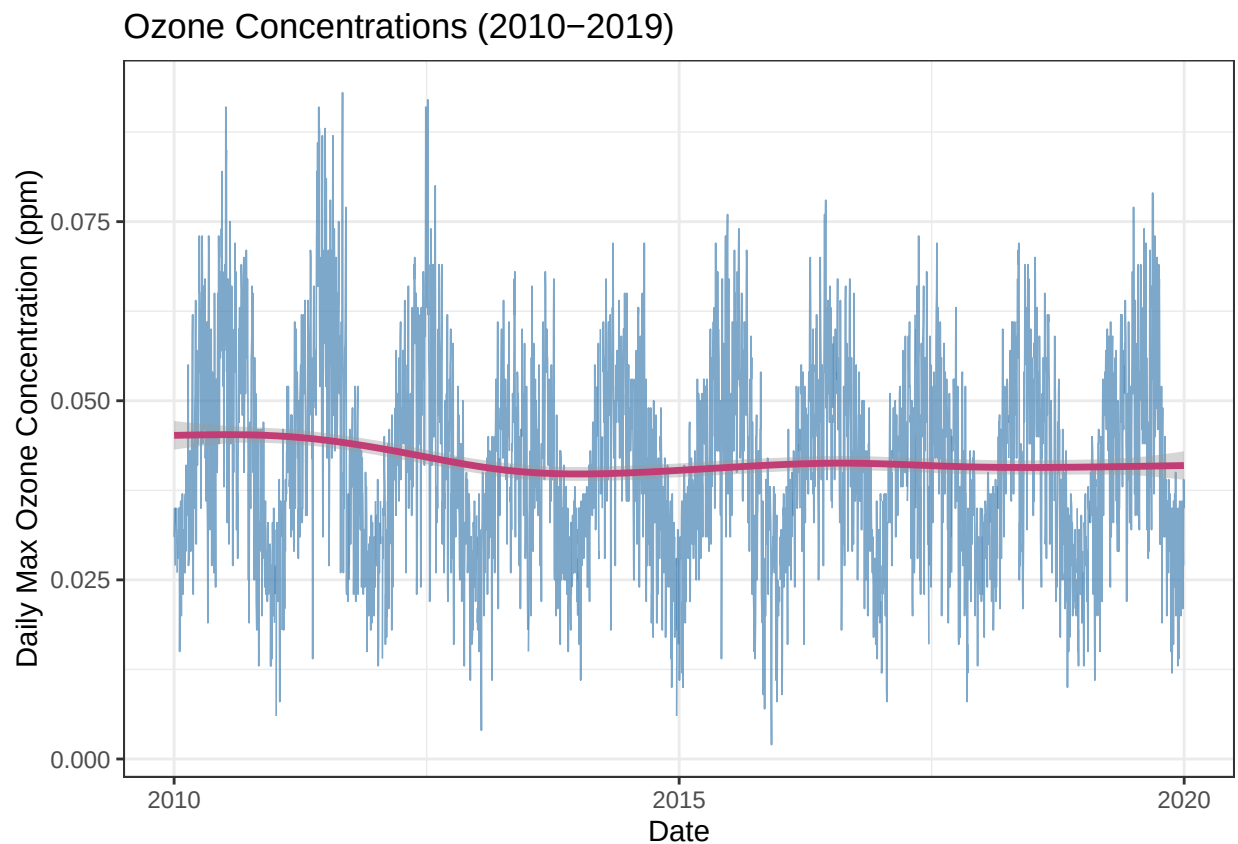
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7  
  
ggplot(GaringerOzone, aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +  
  geom_line(size = 0.3, alpha = 0.7, color = "steelblue") +  
  geom_smooth(method = "gam", color = "#c13d75ff", se = TRUE, size = 1.2) +  
  labs(x = "Date",  
       y = "Daily Max Ozone Concentration (ppm)",  
       title = "Ozone Concentrations (2010-2019)") +  
  theme_bw()
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## 'geom_smooth()' using formula = 'y ~ s(x, bs = "cs")'

## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The plot shows that daily ozone levels fluctuate a lot each year, but overall the concentrations stay fairly balanced through 2010–2019. There are seasonal peaks and dips, I assume because of the four different seasons in the USA, but no strong upward or downward long-term change is obvious from the smoothed line.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <- GaringerOzone %>%
  mutate(Ozone_Interpolated = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration, na.rm = FALSE))

summary(GaringerOzone$Ozone_Interpolated)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: We didn't use a piecewise because linear interpolation shows the data on the daily zone levels that change slowly from one day to the next. A piecewise constant method would create flat lines, which isn't realistic for air quality data that fluctuates daily.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(Year = year(Date),
         Month = month(Date)) %>%
  group_by(Year, Month) %>%
  summarize(Mean_Ozone = mean(Ozone_Interpolated, na.rm = TRUE),
            .groups = "drop") %>%
  mutate(Date = ymd(paste(Year, Month, "01", sep = "-")))

# Check monthly data
head(GaringerOzone.monthly)
```

```
## # A tibble: 6 x 4
##   Year Month Mean_Ozone Date
##   <dbl> <dbl>     <dbl> <date>
## 1  2010     1     0.0305 2010-01-01
## 2  2010     2     0.0345 2010-02-01
## 3  2010     3     0.0446 2010-03-01
## 4  2010     4     0.0556 2010-04-01
## 5  2010     5     0.0466 2010-05-01
## 6  2010     6     0.0576 2010-06-01
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

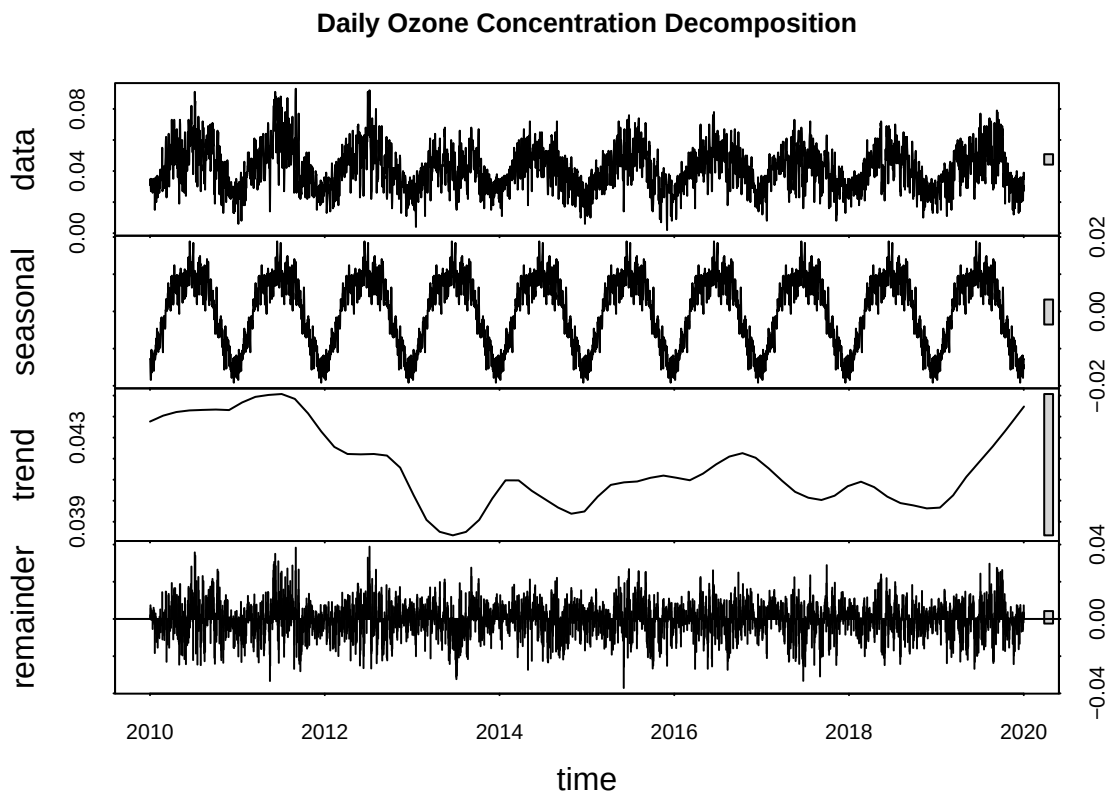
```
#10
GaringerOzone.daily.ts <- ts(GaringerOzone$Ozone_Interpolated,
                             start = c(2010, 1),
                             frequency = 365)

# Monthly time series
```

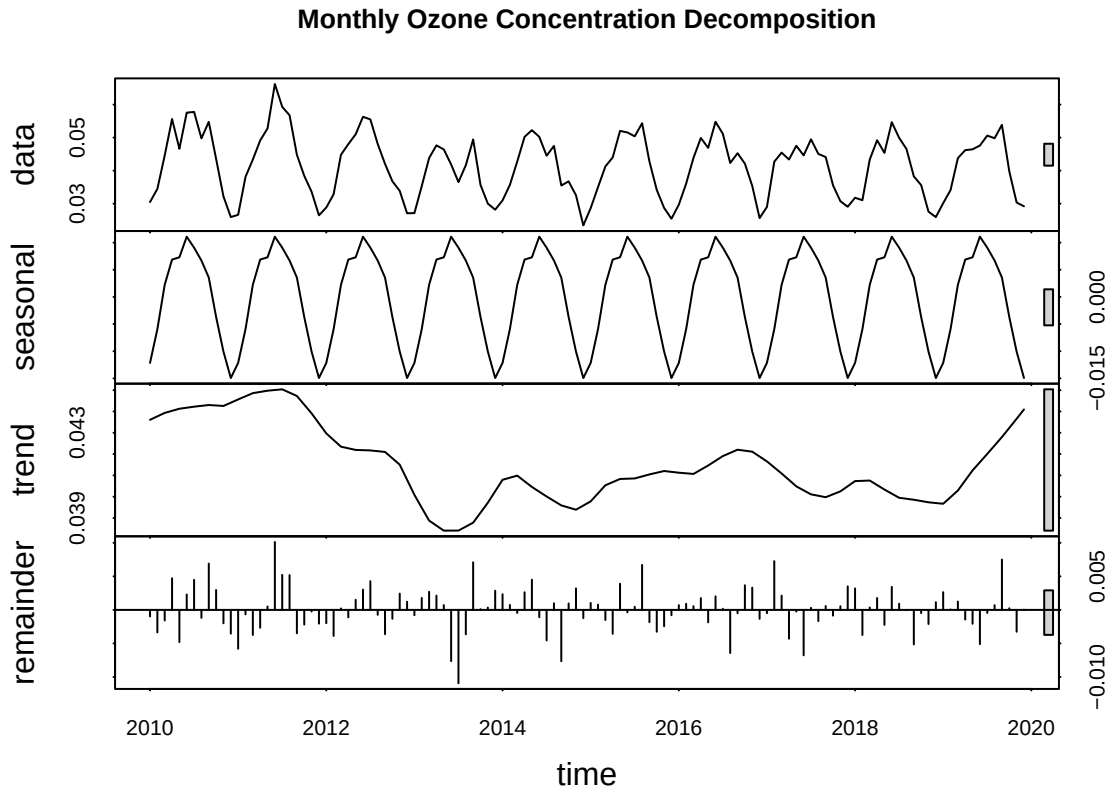
```
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Mean_Ozone,
                              start = c(2010, 1),
                              frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
# Daily decomposition
GaringerOzone.daily.decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(GaringerOzone.daily.decomp, main = "Daily Ozone Concentration Decomposition")
```



```
# Monthly decomposition
GaringerOzone.monthly.decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone.monthly.decomp, main = "Monthly Ozone Concentration Decomposition")
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
ozone_trend <- trend::smk.test(GaringerOzone.monthly.ts)
ozone_trend
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
## S varS
## -77 1499
```

```
summary(ozone_trend)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## alternative hypothesis: two.sided
```

```
##
## Statistics for individual seasons
##
## H0
##
##      S varS    tau      z Pr(>|z|)
## Season 1:  S = 0   15  125  0.333  1.252  0.21050
## Season 2:  S = 0   -1  125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4  124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 5:  S = 0  -15  125 -0.333 -1.252  0.21050
## Season 6:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 7:  S = 0  -11  125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7  125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 11: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 12: S = 0   11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

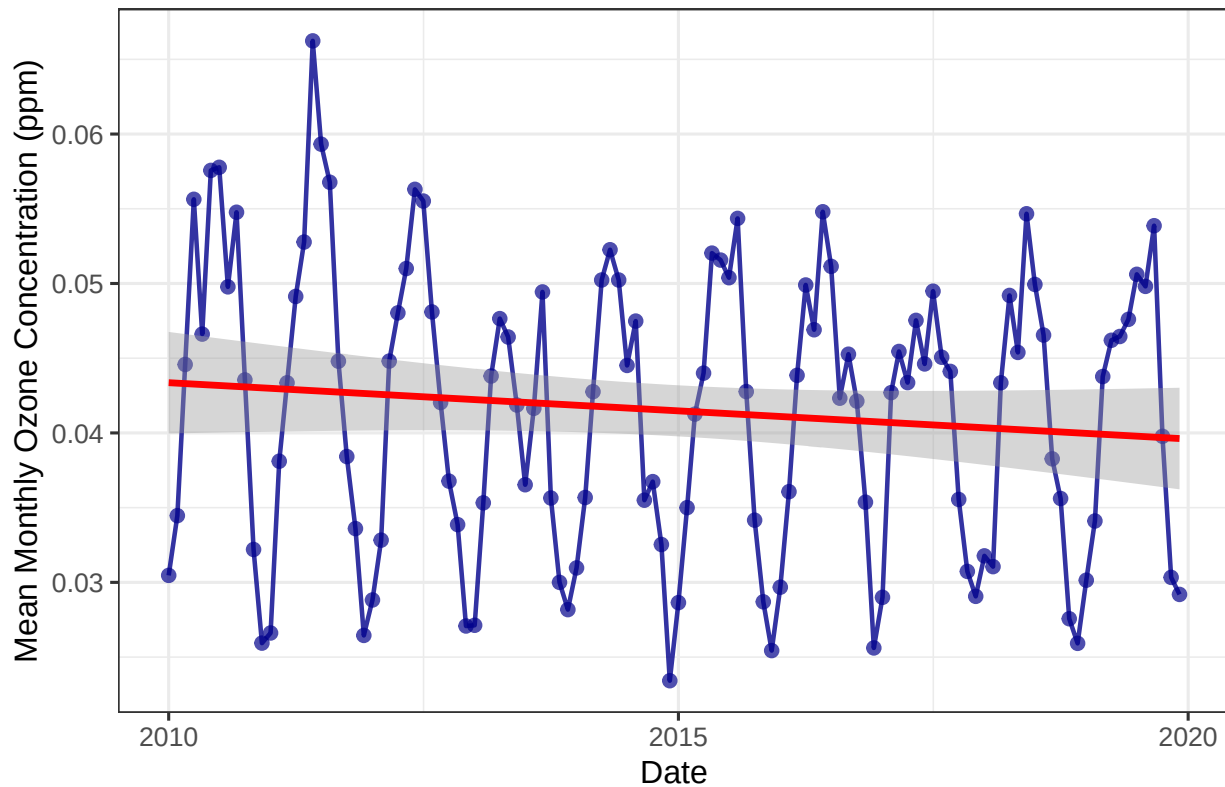
Answer: The Seasonal Mann-Kendall is the appropriate one because the data has a solid, repeating seasonal pattern, per seasons, which is very clear in the decomposition plot and patterns. Furthermore, the seasonal version correctly looks for a trend within each seasons.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y = Mean_Ozone)) +
  geom_point(size = 2, alpha = 0.7, color = "darkblue") +
  geom_line(size = 0.8, alpha = 0.8, color = "darkblue") +
  geom_smooth(method = "lm", color = "red", se = TRUE, size = 1.2) +
  labs(x = "Date",
       y = "Mean Monthly Ozone Concentration (ppm)",
       title = "Monthly Average Ozone Concentrations (2010-2019)") +
  theme_bw(base_size = 12)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Monthly Average Ozone Concentrations (2010–2019)



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Based on the analysis, ozone concentrations at the Garinger High School station showed a small but statistically significant decreasing trend from 2010 to 2019. The plot of monthly averages shows the data fluctuating a lot from year to year, but the overall direction is downward. This is supported by the Seasonal Mann-Kendall test, which found a significant trend with $z = -1.963$, $p\text{-value} = 0.04965$.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

#15

```
GaringerOzone.monthly.nonseas <- GaringerOzone.monthly.ts -
  GaringerOzone.monthly.decomp$time.series[, "seasonal"]
```

#16

```
ozone_nonseas_trend <- trend::mk.test(GaringerOzone.monthly.nonseas)
ozone_nonseas_trend
```

```
##
## Mann-Kendall trend test
##
## data:  GaringerOzone.monthly.nonseas
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.179000e+03  1.943657e+05 -1.651376e-01
```

Answer: The Mann-Kendall test still detects a significant downward trend ($z = -2.67$, $p = 0.0075$). This confirms that the decrease in ozone concentrations is not just a seasonal effect. It's a true long-term decline across years.

AI Statement:

I did not use AI to complete this assignment. However, I used AI support from Grammarly to correct my grammar errors and paragraph responses.